

TrustScientist: Evidence and Branch Revalidation for Autonomous Research Agents: Technical Appendix

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CCS Concepts

• Computing methodologies → Artificial intelligence.

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Research agents, evidence verification, reproducibility

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1 Recorded Case Studies

These four cases were selected after inspecting the saved experiments to illustrate acquisition, version propagation, branch replay, and a binding failure. The complete cohorts remain in the aggregate evaluation. The companion `revision_examples/examples.json` contains the selected records, source selectors, and file hashes. No new model call or human annotation was made for these illustrations. Version and branch updates are controlled interventions; acquisition operates on published artifacts.

1.1 Restoring a reporting state through two evidence actions

Case 1d41a1787bf4 asks for the SVM F1 score on Musk1+. Its source is accessible and intact, but the capsule binding is incorrect and extraction is interrupted. The initial frontier contains binding; extraction cannot proceed until that prerequisite passes.

Step	Executed action	Binding	Extraction
Initial	—	Fail	Unresolved
First	Rebind artifact	Pass	Unresolved
Second	Resume extraction	Pass	Pass

The two actions read the same source, write separate receipts, and change the public evidence digest after each step. Re-audit resolves extraction, quotation, and the semantic rubric. The final binding quotes

Classifier svm: 87.5155 1.9971

The answer 87.5155 matches the benchmark reference. The guard-free controller closes after two actions costing two declared units. The structured selector also produces a valid closure for this paired case. The guarded ECC record has a correct candidate but declares no closure: historical calibration fails its target. This example connects an executed action to new evidence and re-audit; statistical certification remains unresolved.



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1.2 A shared source invalidates several certificates

In capsule 08ec5a4786de2e71, event 1 changes registered source f000. Three reporting chains read that source. Their extraction, binding, and closure nodes are descendants of the same changed input. Figure 1 preserves the recorded dependency structure.

Closure node	Graph cache after re-audit	Stateful LLM cache
n004	Fail	Fail
n007	Fail	Pass (stale)
n010	Fail	Pass (stale)

The stateful Qwen validator rechecks only the first reporting chain and retains the other two closures. Graph and flat dependency tracking both recheck ten affected nodes and retain no stale certificate; full revalidation checks eleven. The unaffected source node keeps its previous value. These counts concern the supplied version contract. They show the effect of executing transitive invalidation in this event; the flat representation provides the same result.

1.3 The first affected decision need not be the rollback point

Capsule f8ff1f0dc2ec860b, event 1, changes a source read by both decisions in an authored history. The earlier decision already selected REFINE. The later decision selected PROCEED under the preceding evidence. Counterfactual replay gives:

Decision	Old action	Replayed action	History operation
Earlier	REFINE	REFINE	Retain receipt
Later	PROCEED	REFINE	Invalidate suffix

CBR refreshes the earlier attestation and rolls back before the later decision. The stored rollback index is 1 in the zero-based archive, and the active history retains one decision. Full replay selects the same checkpoint with the same two decision evaluations in this event. The archived certificate-only tracker marks both certificate predicates false but keeps the historical PROCEED action, leaving one obsolete action. The case illustrates why replay compares actions before choosing a checkpoint. It exercises a controlled deterministic routing rule; no replacement scientific experiment is executed.

1.4 A matching value can still bind to the wrong metric

A nominal-clean manifest in capsule-4180912 asks:

Report the macro f1 score for the random classifier.

The candidate value is 0.4858800853546077. The saved source context assigns that value to the neural network’s *macro recall*. The file exists, the numeric type and path pass, and the observed digest matches the declared digest. The candidate builder also supplies a true relation flag.

TrustScientist accepts those flags and returns CLOSED. The GPT-5.6-Sol response leaves the same manifest OPEN. Its reason identifies both the metric and classifier mismatch. This is a failure of the supplied binding, retained among the nominal-clean examples. The certificate faithfully records the represented checks, while its accepted relation flag leaves the claim unsupported by the quoted context. The example explains why nominal clean closure is reported separately from established semantic correctness.

2 Metrics and Repeated Decisions

Let A_i indicate a closed prediction and let y_i and $\hat{y}_i$ be the gold and predicted relations. The relation vocabulary is SUPPORTS, REFUTES, and NEI. With $N = 339$, $N_{\text{NEI}} = 130$, and $N_{\text{evidence}} = 209$:

$$\begin{aligned} \text{coverage} &= N^{-1} \sum_i A_i, \\ \text{NEI-FCR} &= N_{\text{NEI}}^{-1} \sum_i A_i \mathbf{1}[y_i = \text{NEI}], \\ \text{valid recall} &= N_{\text{evidence}}^{-1} \sum_i \mathbf{1}[y_i \neq \text{NEI}, \hat{y}_i = y_i], \\ \text{selective risk} &= \frac{\sum_i A_i \mathbf{1}[\hat{y}_i \neq y_i]}{\sum_i A_i}. \end{aligned}$$

Macro-F1 averages class F1 after mapping unresolved predictions to NEI. No claim-level artifact certificate is inferred from a support or refutation label. The exact frozen NLI checkpoint is:

```
MoritzLaurer/  
DeBERTa-v3-base-mnli-fever-anli
```

The SciFact LLM endpoint is deepseek-chat, distinct from the CORE manifest baseline. The training/development unit is a claim-document pair; pairs sharing a claim form a dependence cluster. The current bootstrap analysis resamples 300 claim clusters with 5,000 draws. It differs from an older row-bootstrap script retained in the project.

All five saved SciFact runs produce identical operating-point labels for the LLM judge, global consensus, and TrustScientist policies. The sample variance of coverage, NEI-FCR, valid-claim recall, Macro-F1, and selective risk is zero. NLI-only, risk-controlled NLI, and Mondrian policies are deterministic in this evaluation. Repetition still changes some LLM confidence values and therefore their rankings. It does not provide five independent test sets.

The 20 TrustScientist errors among 118 selections decompose into 11 NEI closures and nine reversed SUPPORTS/REFUTES relations. The remaining 98 are correct. Of the 209 evidence-bearing pairs, 111 are not correctly resolved: 102 remain unresolved and nine are assigned the wrong relation. This accounts for both the selective-risk and valid-recall denominators.

2.1 Rationale-hit diagnostic

The frozen selector hits a gold rationale on 118/209 evidence-bearing pairs (56.5%). Table 1 stratifies the saved TrustScientist predictions by that indicator. Rationale misses account for eight of nine wrong selected relations on evidence-bearing pairs. Correct decisions remain possible on some misses because a non-annotated sentence may still convey relevant information. This post hoc stratification does not replace an oracle-retrieval intervention and cannot

Table 1: Post hoc diagnostic on evidence-bearing pairs only.

Rationale	Pairs	Closed	Correct	Wrong
Hit	118	79	78	1
Miss	91	28	20	8

Table 2: Original normalized NEI-FCR integrals. Variance is the sample variance across five saved runs, not a confidence interval.

Policy	Mean	Variance
Direct NLI	0.05040	0
LLM judge	0.06580	0.00001775
Typed NLI	0.04787	0
Global consensus	0.02597	0.000004664
TrustScientist	0.02413	0.000006645

causally assign each error to a retriever or verifier. NEI pairs have no gold rationale and are excluded.

2.2 The original matched-coverage analysis

At each point, the saved procedure retains the top k natural non-NEI closures for each policy. The saved shared interval selects between 17 and 93 of the 339 pairs, using the resulting coverage coordinates. The ordinate is NEI-FCR, not selective risk. The reported area is the trapezoidal integral divided by interval length; it therefore has the scale of a mean NEI-FCR over the interval.

The typed ranking uses $\min(1, s/\tau)$, so every accepted typed-NLI example has score one. Stable sorting leaves these ties in dataset order. On their accepted subsets, the full and global consensus rankings reduce to LLM confidence because the NLI priority is also clipped to one. The resulting areas are reproducible but sensitive to this ranking design; they do not establish superiority over a well-ranked comparator.

3 CORE Manifest Protocol and Baselines

The 79 unique question targets come from all 45 CORE-Bench test capsules. The evaluator searches extracted text for the known reference value and accepts the first match. It assigns the target question to the candidate record and sets its clean relation flag to true. Eighteen of 62 extracted candidates would fail the preceding weak lexical relation heuristic. That heuristic itself has no independent accuracy validation, so 18 is a code-path sensitivity count, not a count of proven wrong bindings. The 62/79 endpoint is reference-assisted candidate retrieval. Seventeen nonextractable targets remain OPEN with action enable_or_extend_artifact_parser.

Both LLMs see the same 372 manifest inputs and text prompt. Reconstructing inputs from the decision table reproduces the saved input JSONL exactly. The prompt asks the model to close only if an admissible artifact satisfies every mandatory condition, to return OPEN for missing/invalid conditions, and UNVERIFIABLE for evidence it cannot inspect reliably. It asks for a JSON state, confidence, failed condition, and reason. The user message includes claim, requirement, type, mandatory flag, artifact manifest, and

Table 3: CLOSED decisions by supplied manifest condition. Nominal clean states have not been independently validated for semantic correctness.

Condition	DeepSeek	GPT	TrustScientist
Nominal clean	55/62	59/62	62/62
Missing artifact	0/62	0/62	0/62
Wrong type	39/62	0/62	0/62
Invalid path	49/62	0/62	0/62
Stale digest	29/62	0/62	0/62
Wrong question	32/62	0/62	0/62
All corrupted	149/310	0/310	0/310

semantic evidence. It does not request four separate type decisions with a shared external aggregator.

The manifest supplies validity flags for artifact type, path, digest, and observed relation. The type is SUFF for nominal clean inputs, AUTH for missing-artifact, wrong-type, invalid-path, and stale-digest controls, and REL for wrong-question controls. It consequently changes with the control condition. Gold states, condition labels, and final TrustScientist decisions are retained for analysis but excluded from the model-facing prompt.

DeepSeek requests deepseek-v4-flash with temperature zero and JSON object output. GPT requests and returns gpt-5.6-sol, with reasoning effort none and a strict JSON response schema. The text prompt and evidence inputs are matched; the model and output protocol differ. No new model calls were made for the present reanalysis.

For DeepSeek, 55 clean-closed cases generate 275 corresponding corruptions. Of those, 148 remain closed (53.8%); another corrupted closure comes from a clean-open case. This explains why conditional preservation is 148/275 while residual corrupted closure is 149/310. GPT has 0/295 conditional preserved closures. The main paper uses the unconditional corrupted denominator so all policies are compared on the same states.

The GPT clean nonclosures expose three mismatches in the saved contexts:

- capsule-4180912: the question requests random-classifier macro-F1; the candidate value instead describes the neural network’s macro recall.
- capsule-8234136: the question requests lowest average wait time, while the candidate appears in a CPU-utilization plot.
- capsule-9832712: the question requests a Figure 2 availability percentage, while the number 100 comes from a bibliography title.

These are evidence of limitations in the candidate-building procedure. Counting all nominal clean closures as correct would reward acceptance of such mismatches.

3.1 Cumulative gates and restoration

The archived cumulative value, type, artifact-binding, and relation stages leave 248, 186, 62, and 0 of the 310 controls closed. These exact multiples follow from five equally sized classes: value checks

Table 4: Recomputed fresh-execution outcomes and summed task wall times.

Run	Strict tasks	Fresh questions	Seconds
1	19/20	58/59	840.6
2	19/20	58/59	745.9
3	19/20	58/59	741.5
4	19/20	58/59	747.7
5	19/20	58/59	754.0
Mean	19/20	58/59	765.9

remove missing-artifact controls, type checks remove wrong-type controls, path/digest checks remove two further classes, and relation checks remove wrong-question controls. This is coverage of the constructed test conditions rather than an effect size on a natural-error distribution.

The action name corresponds to the first failed gate. Restoration calls the same verdict function on the saved nominal clean record; all 310 return CLOSED. No repair command runs in this test. The main trace uses clean row 1 and stale-control row 5 of the saved decision table (one-based data records). The reconstructed certificate contains the claim/requirement ID, artifact path, value, expected and observed digest, passed predicates, and decision. Its provenance record identifies it as reconstructed from the archive.

4 Fresh-Workspace Diagnostic

Each run executes the same 20 public-train capsules in a fresh workspace. Strict success requires a locally produced result within the declared target. The saved run also attempts one official-artifact recovery; that recovery is excluded from strict task success. The 58/59 question count refers to initial fresh-output matches, before recovery. Compatibility fixes were prepared for this host, so the diagnostic does not measure autonomous repair selection.

The retained failure is capsule-4884085: its archived CUDA 10.1 binary is incompatible with the Ampere GPU, and CPU fallback misses the reference target. It remains in every denominator. All 20 tasks’ decisions and 59 question transitions were compared with the aggregate summaries.

5 Exploratory Natural-Claim Preannotation

An archived pilot contains 100 saved claims, 25 from each of four agent-generated papers with available source bundles. A generic artifact adapter infers required evidence from claim text/type and checks available artifact categories. Its saved output differs from the CORE policy operating on supplied predicate flags; the two experiments must not be pooled.

GPT-5.6-Sol preannotations were saved separately for each paper, withholding previous judgment values. Recounting the per-claim JSONL files gives 100 adapter CLOSED decisions versus four model CLOSED, 51 OPEN, and 45 UNVERIFIABLE decisions. The saved comparison CSV agrees with every record, and the source and input hashes match the archived snapshot. Model closure counts by paper are 0, 3, 0, and 1 (25 claims each); these are descriptive counts without a correctness reference or a population estimate.

The sample favors introductory and related-work claims; cited external sources are often outside the local bundles. Inputs retain paper identities and provenance cues, so withholding previous judgments does not establish independent blinding or matched evidence access. Each of the two human annotation packets contains 100 records but no populated eligibility, requirement, or claim-state fields. The adjudication table also has no completed human states.

Neither model output nor adapter output is gold. We report no requirement recall or natural false-closure rate from this pilot. An independent evaluation needs a frozen evidence boundary, eligible claim units, and adjudicated requirement identities before one-to-one scoring.

6 Evidence Archive and Assistance Disclosure

The reanalysis checks tables, reconstructs inputs, compares predictions, replays predicates and restoration, and recomputes SciFact and execution metrics. Derived JSON records input hashes using SHA-256. This verifies saved records without reopening original capsule binaries or establishing the correctness of supplied relation flags. The exact manifest-auditor prompt is archived separately.

AI assistance through Codex was used on 25–30 September 2026 to revise the manuscript, inspect code and saved outputs, verify literature metadata, and write the offline reanalysis. The original figures were retained. The session identified its model family as GPT-6; a precise backend revision was not available. The author’s revision brief requested an AAMAS research-agent framing, omission risk, consistent semantic metrics, scoped corruption and action claims, updated related work, raw-output verification, and a worked trace. This September 30 request asked to update the paper, assess effectiveness, and run additional experiments and revise Overleaf without new human annotation. Codex implemented the paired semantic-gate design, independent scoring code, and manuscript revisions under that brief. A subsequent request to test stateful graphs against carefully prompted validators led to the dynamic file-version experiment, its dependency-table control, and independent replay scorer; that protocol and the exact model prompt are archived. The new experiment uses actual model calls; the separate calibration construction uses stated assumptions. A further request led to the frozen discriminating-acquisition pilot and a separately labelled post hoc choice-interface diagnostic on newly acquired public-train capsules; their prompts, source hashes and model receipts are archived. Experimental model runs and their saved prompts are documented separately; AI-generated labels are not described as human or expert validation.

7 Archived Restoration Trace

7.1 Worked trace: a stale artifact manifest

The archived control for capsule-5507257 follows one requirement through the interface. Its task asks for the multitask model’s final test accuracy. Reference-assisted extraction locates 96.12499 (rounded) in results/output. The represented checks require presence, an accepted type, a permitted path, matching digest, and the declared question relation. These checks cover this benchmark target; the trace does not create replication or comparison requirements absent from the target.

The stale control retains the value and context but replaces the observed digest with 64 zeroes. The AUTH predicate comparing observed and expected digests fails, leaving the requirement OPEN. The saved action specification is refresh_stale_artifact, linked to the originating requirement and its digest test. For a real executor, the expected output would be a refreshed artifact record whose digest can be checked again.

In the archived experiment, the intervention restores the known clean manifest. Its digest again matches the expected value beginning bf0c92e1. Re-audit repeats the same checks and returns CLOSED. A reconstructed certificate view contains the requirement identifier ending 4f0c98b26e, the artifact reference, extracted value, digest comparison, predicate outcomes, and final state. The underlying records support this view, but no originally issued certificate

file is archived. The sequence shows claim → requirement → candidate binding → failed predicate → action specification → changed manifest → re-audit → certificate view. The actual intervention is manifest restoration, not execution of the proposed refresh action.

8 Controller Protocol and Historical Evidence

The control experiment freezes the controller, adapter, case list, calibration split, and source hashes before full evaluation. Each case is evaluated under six policies and budgets 1, 2, and 4. The local Qwen3-8B endpoint uses temperature zero, seed 20260929, non-thinking JSON mode, and at most 512 output tokens per request. Identical request payloads share cached responses across policies and budgets. Rollouts are policy executions on shared model outputs, not independent model replications. Tool executions still produce separate receipts. Capsule-cluster percentile intervals use 5,000 bootstrap resamples; degenerate zero-error intervals do not establish a population risk bound.

Inference uses vLLM 0.19.1, bfloat16, and a 16,384-token context window. The Qwen3-8B weight snapshot is b968826d9c46dd6066d109eabc6255188de91218. At budget 4, mean acquisition cost over all natural attempts is 1.83 units for ECC and 1.70 for the structured selector; on controlled attempts it is 1.67 and 1.89, respectively. Costs of unsuccessful attempts are included. The corresponding mean charged token counts are 10,840 versus 13,385 on natural tasks and 1,959 versus 2,112 on controlled states. They include shared cached requests; actual unique inference consumes 818,472 tokens.

Graph dependencies are access → integrity and binding; integrity and binding → extraction; extraction → quotation → semantic adequacy. The cost-aware score counts failed nodes in a fixed tool-effect map and divides by declared cost. Refresh and fetch can affect four structural nodes; rebind affects binding, resume affects extraction, and search affects quotation and semantic adequacy. The graph-free order is resume, rebind, search, fetch, refresh. The budget-selection ablation uses refresh, fetch, search, rebind, resume. Exact model prompts, decisions, action receipts, and evidence hashes are retained with the experiment release. This is an authored scheduling heuristic, not a learned value-of-information estimator.

8.1 Earlier formal artifact protocol

We preserve the earlier 1,371-record protocol separately: 270 requirement records, 864 controlled attempts, and 237 natural-case model runs. Its 123 natural failure trajectories include abstentions and protocol failures; they are not 123 independently annotated false closures. The controlled cohort has 24 nominal-clean bundles from 11 capsules and five corruption variants each. It uses the same local model family but a different policy, input and scoring protocol from the new graph experiment.

The earlier headline percentages 68.06%, 67.36%, 0.69%, and 2.78% describe *error incidence*, not risk conditional on closure. Both recovery rounds recover 69/120 corrupted states (57.50%); the second round adds no successes. False recovery is 3/72 declared recoveries (4.17%). The executor restores published source artifacts; these are not autonomous scientific repairs.

Table 5: Earlier controlled protocol, 144 attempts per policy. Error incidence uses all attempts; conditional risk uses closed attempts.

Policy	Errors/all	Errors/closed	Valid/all
Direct	98/144	98/121	23/144
ReAct	97/144	97/120	23/144
ReAct + Reflection	97/144	97/120	23/144
LLM Judge	97/144	97/120	23/144
TrustScientist	1/144	1/24	23/144
TrustScientist + loop	4/144	4/96	92/144

In a post hoc replay, applying the same deterministic checks to the saved Direct, ReAct, Reflection, and Judge reports yields 24 closures, one error, and 23 correct closures for each method. All 144 decisions agree with the static TrustScientist policy. These 576 replay records use no new model calls and show that the earlier gap cannot be attributed to typing alone.

8.2 Additional fresh execution diagnostic

A separate new execution of the same 20-task, 59-question subset initially reproduces 12/20 tasks and 40/59 questions. Operator-selected environment repairs and six reruns increase this to 18/20 and 56/59. Extending the output parser for scalar wrappers, without changing the raw scientific output, produces 19/20 and 58/59. This sequence is retained separately from the five archived runs and the autonomous artifact-acquisition loop. It does not show autonomous selection of environment repairs. The remaining reference mismatch stays in the denominator.

Table 6: All control settings. V and F are valid and false final closures; Rec is valid recovery among initially unresolved states. A and K are mean actions and declared cost among valid final closures; a dash means undefined. Candidate outcomes are shown in the main text. Mean tokens charge shared cached calls to each policy, and do not count independent inference.

Cohort	Policy	Budget	V/all	F/all	Rec	A	K	Mean tokens
Natural	ECC	1	0/46	0/46	0/24	–	–	5483
Natural	ECC	2	0/46	0/46	0/24	–	–	8752
Natural	ECC	4	0/46	0/46	0/24	–	–	10840
Natural	w/o graph	1	0/46	0/46	0/24	–	–	5483
Natural	w/o graph	2	0/46	0/46	0/24	–	–	8752
Natural	w/o graph	4	0/46	0/46	0/24	–	–	10840
Natural	w/o risk guard	1	18/46	4/46	0/24	0.00	0.00	5483
Natural	w/o risk guard	2	21/46	7/46	3/24	0.14	0.29	8752
Natural	w/o risk guard	4	21/46	9/46	3/24	0.14	0.29	10840
Natural	w/o budget selection	1	0/46	0/46	0/24	–	–	5483
Natural	w/o budget selection	2	0/46	0/46	0/24	–	–	8752
Natural	w/o budget selection	4	0/46	0/46	0/24	–	–	10840
Natural	w/o re-audit	1	0/46	0/46	0/24	–	–	5483
Natural	w/o re-audit	2	0/46	0/46	0/24	–	–	5483
Natural	w/o re-audit	4	0/46	0/46	0/24	–	–	5483
Natural	Structured LLM	1	18/46	4/46	0/24	0.00	0.00	5605
Natural	Structured LLM	2	21/46	7/46	3/24	0.14	0.29	10245
Natural	Structured LLM	4	21/46	9/46	3/24	0.14	0.29	13385
Controlled	ECC	1	0/36	0/36	0/24	–	–	654
Controlled	ECC	2	0/36	0/36	0/24	–	–	1303
Controlled	ECC	4	0/36	0/36	0/24	–	–	1959
Controlled	w/o graph	1	0/36	0/36	0/24	–	–	654
Controlled	w/o graph	2	0/36	0/36	0/24	–	–	1303
Controlled	w/o graph	4	0/36	0/36	0/24	–	–	1303
Controlled	w/o risk guard	1	0/36	0/36	0/24	–	–	654
Controlled	w/o risk guard	2	11/36	1/36	11/24	2.00	2.00	1303
Controlled	w/o risk guard	4	22/36	2/36	22/24	1.50	2.50	1959
Controlled	w/o budget selection	1	0/36	0/36	0/24	–	–	654
Controlled	w/o budget selection	2	0/36	0/36	0/24	–	–	654
Controlled	w/o budget selection	4	0/36	0/36	0/24	–	–	1967
Controlled	w/o re-audit	1	0/36	0/36	0/24	–	–	654
Controlled	w/o re-audit	2	0/36	0/36	0/24	–	–	654
Controlled	w/o re-audit	4	0/36	0/36	0/24	–	–	654
Controlled	Structured LLM	1	0/36	0/36	0/24	–	–	880
Controlled	Structured LLM	2	10/36	1/36	10/24	2.00	2.00	1463
Controlled	Structured LLM	4	22/36	2/36	22/24	1.18	2.82	2112

9 Operational Labels and Policy Certification

9.1 What the existing labels establish

This study adds no human-annotation experiment. CORE questions and reference answers supply an existing external task specification. In the control run, the final evaluator conjuncts the same structural and literal-binding checks used by the controller with reference-answer matching. Withholding references from action selection prevents direct answer access; it does not make the shared checking implementation an independent validator. The reported counts retain this frozen operational definition.

A retrospective paired audit matches task identifiers across the ungaurded controller and structured selector at budget 4. All 46 natural and 36 controlled cases agree on both the closure decision and operational validity. The audit does not assert identical generated text or identical action sequences.

Source fidelity and reference agreement can diverge. For a question about unavailable analysis scripts, a saved report quotes a row for processed data. For a sample-size question, another quotes a regression-estimate row. These are observable category mismatches. A separate classification report quotes an output value that differs from the benchmark reference executions. That disagreement alone cannot establish a wrong quotation or distinguish execution variation from a context error. Plot-based answers can also lose spatial relations when figures are flattened into text. These examples motivate separate scoring of task-answer agreement, source-report fidelity, and execution or artifact-contract validity. The existing aggregate labels have not been retrospectively replaced with inferred semantic gold.

9.2 A sufficient episode-level calibration construction

The following construction specifies a route to certification; it is not a guarantee attained by the historical guard. Fix a finite family of M complete episode policies before calibration, including their prompts, tools, budgets, internal stopping rules, and model randomization distributions. Each policy may acquire evidence adaptively within an episode. Fix an operational binary error evaluator and a calibration sample size before observing its outcomes. Calibration episodes must be independent and identically distributed from the deployment population and unused in policy or evaluator development. Episodes must not share adaptive memory. A fixed benchmark collection alone does not establish this population-sampling assumption.

For policy j , let C_j indicate final closure and L_j indicate evaluator error. Its target is $r_j = \Pr(L_j = 1 \mid C_j = 1)$, for positive closure probability. Among the calibration episodes let N_j close, with K_j errors. For a fixed policy, conditional on N_j , the accepted-error count is binomial with parameter r_j . Define the one-sided Clopper–Pearson upper bound

$$U_j = \begin{cases} 1, & N_j = 0 \text{ or } K_j = N_j, \\ Q_{1-\delta/M}(\text{Beta}(K_j + 1, N_j - K_j)), & \text{otherwise,} \end{cases} \quad (1)$$

where Q_p denotes a distribution’s p -quantile. Certify only policies with $N_j > 0$ and $U_j \leq \alpha$, for a prespecified tolerance $0 < \alpha < 1$.

With probability at least $1 - \delta$ over calibration,

$$\forall j \text{ certified, } r_j \leq \alpha. \quad (2)$$

To see this, the exact binomial bound has undercoverage probability at most δ/M for each fixed policy, also after averaging over N_j . A union bound over the M policies gives simultaneous coverage. Any selection among the certified policies preserves this statement. The policies’ outcomes may be correlated on shared episodes; independence is required across sampled episodes, not across policies. This is a standard finite-family calibration argument, consistent with Learn then Test, not a new risk-control theorem.

Internal adaptive acquisition is part of the frozen policy’s output map. The construction neither bounds risk conditional on every observed history nor permits changing the policy after calibration. It does not allow repeatedly inspecting results and stopping calibration at the first favorable bound. If several questions share a capsule, one can prespecify one question per independent capsule, or define a capsule-level any-error loss; the latter certifies a different, capsule-level target. Replicate seeds and controlled mutations are not independent capsules. The guarantee concerns the chosen operational error, and does not repair erroneous labels or omitted obligations.

Even with zero accepted errors, the bound is $U_j = 1 - (\delta/M)^{1/N_j}$. At $\delta = 0.05$, a fixed policy needs at least 59 accepted independent units to certify $\alpha = 0.05$, or 29 to certify $\alpha = 0.10$. A family of ten policies needs 104 or 51, respectively. These are mathematical best-case requirements, not achieved sample counts. Nonzero errors require more evidence. The current historical calibration and previously inspected evaluation cases do not provide this new policy-matched calibration sample. No policy is certified by this construction in our results.

10 Paired Semantic-Gate Protocol

The acquisition failures motivate a narrower test: once a report and its evidence are fixed, does checking the requested scope improve the closure decision beyond structural checks and answer agreement?

10.1 Paired design and independent scoring

We use all 79 existing natural reporting questions. These questions were available during development; this is an exploratory comparison, not an untouched test set. A question-only retrieval preserves table rows and local headings. Each question receives one Qwen3-8B report, one structured rubric review, and one blind second answer with scope witnesses. The second answer does not see the proposed report or reference answer. All decision rules reuse these same outputs, making the comparison paired at the report level.

The common contract checks artifact structure, literal answer support, the represented question, and declared completion. A structured-rubric gate adds the six semantic checks used in the acquisition study. An agreement gate instead requires agreement with the second answer, grounded candidate values, and no declared ambiguity. The scope gate adds literal question anchors and evidence quotations for the requested dimensions, including metric and figure identifiers when present. Comparing the latter two removes only the scope conditions; retrieval, prompts, answers, and all other checks stay fixed. The gate checks a fixed metric vocabulary and figure markers; other dimensions depend on the model’s proposed inventory. Literal witnesses expose a relation for inspection but establish neither its truth nor inventory completeness.

A separately implemented evaluator scores agreement with the benchmark reference values, without calling the controller’s audit or binding functions. It independently checks source-file digests and literal answer support as separate diagnostics. Semantic source fidelity remains unlabelled. Correct and wrong closures below therefore refer to reference agreement; they do not certify unrestricted scientific validity. Unresolved and malformed replies remain in the denominator. Capsule resampling gives paired uncertainty intervals, preserving within-capsule dependence.

10.2 Execution and uncertainty

The experiment uses 45 capsules and 237 actual model requests, with no transport errors. Six replies fail JSON parsing and remain unresolved. The saved responses account for 1,044,094 tokens across the three requests per question. Temperature is zero, the seed is 20270928, the maximum output is 1,536 tokens, and the server context limit is 32,768 tokens. The Qwen3-8B checkpoint is the same as in the control study. Report generation and the blind witness receive identical public evidence bundles; the rubric review also receives the proposed report. No model receives benchmark reference values or the private task identifiers.

Each gate can be evaluated from the shared report and at most one additional model output. The three-call study cost is not the deployment cost of each gate, and no cost advantage is inferred from this replay design. There are no acquisition rounds after the initial retrieval; this comparison isolates a termination decision rather than an autonomous recovery policy.

The separately implemented evaluator treats a numeric answer as matching when it lies within the benchmark reference range, extended by the maximum of 10^{-5} , 5% of the range width, and 0.1% of the absolute lower endpoint. Text answers use normalized exact matching. This preserves the existing operational scoring tolerance without calling its implementation. Reference agreement and literal support do not establish correct row/column semantics. All 79 source-integrity checks pass; 55 reports have literal answer support. Semantic fidelity remains unknown for all questions under independent labels.

We resample the 45 capsules with replacement 10,000 times and compute each rate after pooling the sampled questions. The same draws are used for paired differences. Scope witnesses minus answer agreement yield a successful-closure difference of -11.4 percentage points (95% interval $[-19.8, -2.9]$) and a false-termination difference of -1.3 points ($[-4.2, 0.0]$). Relative to the structured rubric, the corresponding differences are -12.7 points ($[-23.7, -2.9]$) and $+1.3$ points ($[-4.2, 6.9]$). These exploratory intervals describe the observed capsule collection; they do not repair prior use of the tasks during development or provide a deployment risk certificate.

Validation checks the frozen experiment hash, completeness and uniqueness of cases, three requests per question, public-input equality, absence of reference fields and private identifiers in requests, and nesting of scope acceptance within agreement acceptance. Evaluator regression cases include a valid source, a wrong answer, altered extracted text, and a changed source file. The code, protocol, per-question gate decisions, and full responses are retained.

10.3 Complete paired-gate results

We isolate the remaining semantic decision using all 79 existing natural questions from 45 capsules. Each question receives one context-preserving, question-only retrieval, a Qwen3-8B report, a structured rubric review, and a blind second answer with scope witnesses: 237 actual model requests. The questions were used during development, so this is an exploratory study. All gates reuse the same evidence and model outputs.

The common contract checks structure, literal answer support, the represented question, and declared completion. The rubric adds the same six semantic checks used by the controller. Answer agreement instead adds a blind second answer, grounded candidate values, and no declared ambiguity. Scope witnesses add literal question anchors and evidence quotations for requested dimensions, with mandatory metric and figure checks. The agreement–scope contrast removes only those scope conditions; it is a paired component ablation. Scope completeness and the truth of a quoted relation are not established by these syntactic checks.

A separate evaluator scores benchmark-reference agreement without invoking controller functions. Source-file integrity and literal support are checked independently and kept separate from the primary answer score. Thus a wrong closure below means reference disagreement, not necessarily an unfaithful quotation. Malformed replies remain unresolved in the denominator; there is no independent semantic annotation. Full protocol and scoring tolerances appear in the appendix.

Table 7: Paired gates on 79 natural questions. Correct/wrong refer to benchmark answers. Risk is wrong/closed; all methods share generated outputs.

Gate	Correct	Wrong	Risk (%)
Literal contract	44	11	20.0
Structured rubric	35	4	10.3
Answer agreement	34	6	15.0
Agreement + scope	25	5	16.7

Adding scope witnesses rejects one reference error but loses nine correct closures. The paired capsule-bootstrap difference in successful closure is -11.4 percentage points (95% interval $[-19.8, -2.9]$); the difference in false termination is -1.3 points ($[-4.2, 0.0]$). Selective error increases from 15.0% to 16.7%. The structured rubric has the strongest observed balance among the semantic gates, with 35 correct closures and four errors. The scope extension is therefore not adopted as an improvement to the controller. This result locates a limit of inspectable certificates: requiring more literal witnesses can discard valid answers without resolving semantic misbinding. The experiment tests a gate, not an action-selection advantage.

11 Dynamic Version-Contract Protocol

11.1 Inputs, checks, and fixed event sequence

The inputs are the saved public evidence bundles and reporting outputs of the 79-question paired-gate study. We copy 124 source files and generate separate receipt and report records, preserving 45 capsule boundaries. Identical source paths share one source node within a capsule. Each question adds extraction, binding, and closure nodes; there are 361 nodes in total. A check passes when its registered file digests and parent checks pass. Closure additionally requires the archived initial literal-contract decision. The 55 initially passing contracts are not independently verified scientific answers.

The fixed sequence contains nine events: a change to an unrelated file; changes and restorations of a source, receipt, and report; and a simultaneous source and receipt change followed by restoration of both files. Targets follow lexical order, without selecting for maximum sharing. Updates append a fixed marker to isolated copies; restoration copies the original bytes back. Receipt and report files are checked against registered digests, so these updates test version mismatch, not successful parsing of corrupted JSON. The source and receipt in the multi-input event are distinct files.

All policies share a topological executor and a budget equal to the capsule’s node count per event. Selected nodes are first invalidated, then checked in topological order. A failed parent propagates failure without reading child files. Every selected predicate evaluation counts as a check, including such short circuits. The graph index is constructed once per capsule. The flat implementation seeds direct input matches and scans parent lists to a fixed point. Neither policy asks an LLM to determine the changed-file dependency closure. Full re-audit also skips changes outside the registered input set.

The independent evaluator imports no graph or checker code. It reconstructs every event’s bytes from immutable originals, verifies the recorded digests, and evaluates direct certificate-to-file contracts. Original source files are checked for byte identity after the experiment. A smoke trajectory on the largest graph passes these checks before the formal study. Its nine model requests are separate from the 405 formal requests; no policy adjustment is made after the smoke run.

11.2 Prompted stateful comparator

Qwen3-8B uses the existing checkpoint, temperature zero, seed 20260930, and a 768-token output cap. Each request includes checks in topological order, their direct files and parent identities, the changed-file identifiers, persistent predicate cache, full previous event history, and common budget. Event types, mutation/restore labels, reference answers, and oracle validity are withheld. All 405 formal responses produce valid plans; no fallback is invoked. They consume 232,958 actual tokens. This is an invalidation-planning test; it does not ask the model to reinterpret scientific artifacts.

The system prompt directs the model to start with every check whose direct inputs changed, repeatedly add all dependents until reaching a fixed point, include intermediate and closure nodes even if already false, propagate restorations, skip unregistered files, and return only an invalidation list. Malformed output conservatively selects full revalidation. The complete prompt is retained in the frozen protocol. This gives the comparator the same dependency information and the intended algorithm, not a generic “check the

Table 8: Dynamic version-contract tests. Stale: stale certificates accepted; valid: valid certificates retained. Checks exclude common initialization.

Policy	Stale / 163	Valid / 332	Checks
Full re-audit	0	332	2,888
Stateful LLM	38	327	1,624
Dependency graph	0	332	1,530
Flat dependency table	0	332	1,530
No re-audit	163	332	0

answer” instruction. Tool-mediated execution of that algorithm is represented by the flat-table comparator.

11.3 Denominators, uncertainty, and costs

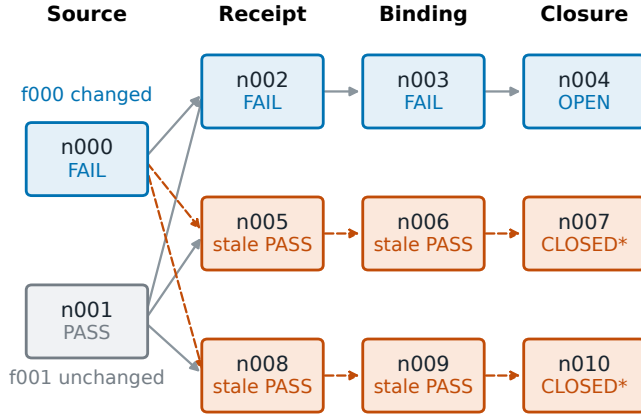
The 79 questions over nine events yield 711 certificate decisions per method. Of these, 216 belong to initially non-passing contracts. The remaining 495 comprise 332 valid and 163 stale contract instances. Counting all invalid instances as the stale denominator would dilute the prompted validator’s error rate from 38/163 to 38/379. We use the former. The valid-retention count includes preserved and restored certificates; it is not recovery from a scientifically wrong answer.

We resample capsules 10,000 times, retaining all events and questions within each sampled capsule. The LLM stale-acceptance interval is [12.8, 35.2]%; its valid-retention interval is [96.2, 100.0]%. The graph’s check reduction relative to the LLM is 5.8% with interval [−7.4, 13.8]%, so no check-cost advantage over that selector is established. Zero observed errors give a degenerate empirical bootstrap interval; this is not a population risk certificate.

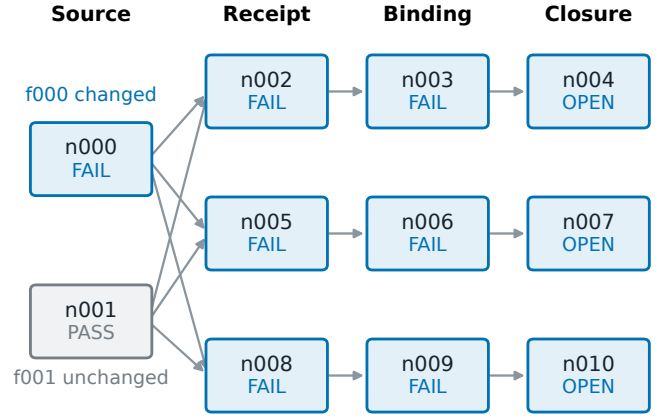
Full re-audit reads 1,915 files and 244,850,308 bytes across the formal events; graph and flat tracking each read 709 files and 70,096,968 bytes. The LLM reads 862 files and 43,905,679 bytes, partly because it misses required revalidation. Its smaller byte count therefore cannot be read as safe efficiency. Common initialization is reported separately in the raw records.

Accumulated checker time is 0.500 seconds for full re-audit, 0.157 for the graph, and 0.156 for the flat table. Graph index construction adds 0.000564 seconds. The LLM’s accumulated request time is 297.184 seconds across concurrent trajectories. These sums are not job elapsed time. The shared server uses a warm file cache and randomized policy execution order; the small deterministic timings are diagnostics, not evidence of a general wall-clock speedup.

(a) Prompted stateful validator



(b) Executable dependency tracking



 Rechecked: FAIL / OPEN
 Cached: stale PASS / CLOSED*
 Unchanged: PASS

Figure 1: An executed shared-source update (capsule 08ec5a4786de2e71, event 1). Both policies receive the same dependencies and changed-file ID. The prompted validator rechecks one of three affected reporting chains, leaving two cached closures stale (CLOSED*). Executable dependency tracking rechecks all affected nodes. The flat-table implementation produces the same right-hand result. Edges and states come from the recorded run; failure means a version-contract mismatch, not a scientifically false answer.

11.4 Executed shared-source failure

Figure 1 shows event 1 in capsule 08ec5a4786de2e71, which updates source f000. Ten checks require revalidation: n000 and n002 through n010. The prompted validator selects only n000, n002, n003, and n004. It follows one downstream chain while omitting two others; cached closures n007 and n010 remain true despite stale source versions. Both deterministic trackers invalidate all ten checks and match the full oracle. Across events, the LLM has 227 exact invalidation sets, 93 sets missing required checks, and 85 with extra checks. The latter two categories need not be treated as disjoint by definition. Seventeen stale acceptances occur on single-source updates and 21 on simultaneous source–receipt updates. This trace demonstrates missing propagation; it does not establish an error rate for naturally occurring artifact changes.

12 Discriminating Evidence Acquisition

12.1 New acquisition cohort and frozen protocol

This study tests whether selecting checks to distinguish competing evidence interpretations improves artifact reporting. The cohort is disjoint from the previously inspected 45 test capsules and 20 cached train capsules. We consider all 25 remaining capsules in the official public training manifest, using an outcome-blind acquisition plan. One archive exceeds the fixed 512-MiB download cap; the other 24 yield 41 questions. All acquired questions remain in the denominator. These are newly acquired evaluation inputs, not the benchmark’s hidden test set.

The tools read published outputs, producing code, metadata, and reproduction instructions. Archives are copied into an isolated release; their code is never executed. Parsing preserves table location and notebook cells. Files larger than 10 MiB, unsupported binaries, and image-only content are not interpreted; parsed text is capped at 400,000 characters per file. The readable collection contains 3,005 documents. Every capsule has some readable output, which does not imply that every requested answer is recoverable.

Two previously inspected capsules supply the operational smoke test. The runner and formal input hashes are frozen before formal model calls; analysis choices are recorded during execution before inspecting outcomes. No policy changes follow formal results. All methods use Qwen3-8B, temperature zero, seed 20260930, and a 2,048-token output limit. Initial retrieval supplies two output passages selected from the question alone. One shared model response proposes an answer, competing interpretations, and possible inspections. Reference answers are never included in these requests.

12.2 Shared state and competing selectors

The common model prompt maintains up to three interpretations h_i with nonnegative planning weights w_i and proposes up to six next probes. For each probe a , it predicts a discrete observable outcome $z_i(a)$ under each interpretation, or marks the prediction unknown. These are heuristic forecasts, not calibrated probabilities or observations. The tested selector uses

$$u(a) = \sum_{(i,j) \in \mathcal{P}(a)} \bar{w}_i \bar{w}_j, \quad a^* = \arg \max_a \frac{u(a)}{c(a)},$$

where $\mathcal{P}(a)$ contains pairs $i < j$ with known, different predicted outcomes. Here $\bar{w}_i$ are normalized positive weights and $c(a)$ is the declared tool cost. Ties use the shared model recommendation, then lower cost. An empty or non-discriminating forecast does not establish information gain.

The stateful validator receives the same hypotheses, probe proposals, files, accumulated observations, previous answer, and action history. It chooses the common model’s recommended affordable probe. Other policies maximize $u(a)$ without dividing by cost, choose the cheapest probe, or choose a seeded random probe. All use the same subsequent model call to update the answer and state. Thus the main contrast changes selection within a model-proposed probe set; it is not a comparison to unrestricted agents or frontier models. The candidate set may itself omit the right explanation.

Output, code, and metadata searches each cost one unit and return at most 3,000 characters; expanding an observed passage costs two units and returns at most 6,000. Each policy has four units and

Table 9: New acquisition cohort, all 41 questions. Correct and wrong count closed answers; costs sum declared tool units over the cohort. The all-initial-probes diagnostic has a different budget and calls the answer model once after all probes.

Selector	Correct	Wrong	Cost
Initial report	13	9	0
Stateful validator	14	11	106
Disagreement / cost	14	11	109
Disagreement only	14	11	108
Cheapest probe	12	10	116
Random probe	14	10	111
All initial probes	14	10	124

continues until the budget is exhausted or no valid affordable unexecuted probe remains. This fixed-budget audit isolates acquisition choices; it does not validate optimal stopping. The all-initial-probes diagnostic executes the entire initial probe bank and then revises once, so it is not a matched four-unit comparator.

12.3 Scoring and implementation checks

The common gate requires a non-null answer, a model completion decision, and an actual quoted output passage. Code alone cannot certify an observed result. The gate does not independently establish that a quotation supports the requested semantic scope. A separate scorer compares answers with the benchmark references using the preexisting numeric tolerance and normalized string matching. Scientific semantic fidelity remains unlabelled.

Successful closure and false termination divide correct and incorrect closed answers by all 41 questions. Recovery counts initially non-null reference-wrong answers that become correct closures. Damage counts initially reference-correct answers that become incorrect closures; loss of a correct closure also includes abstention. Shared initial calls are charged to each policy for per-policy cost but counted once in total actual requests. Tokens, returned characters, actions, and tool units are reported separately. Prefix observations are not independently optimized experiments at smaller budgets.

The primary contrast is disagreement-per-cost versus the stateful validator at the audit endpoint. We resample capsules 10,000 times and retain all questions and methods within each sampled capsule. Secondary comparisons are exploratory. No lower false-termination count alone establishes an improvement, and a zero-error bootstrap interval is not a risk guarantee. Independent implementation checks replay recorded text spans, source hashes, literal gates, budgets, input allowlists, and unique response IDs. All parse and transport failures remain in the evaluation.

12.4 Frozen pilot outcomes and mechanism diagnosis

The frozen study produces 623 actual requests with 2,867,307 tokens, no transport failures, and no malformed JSON responses. The separate smoke test uses 35 requests. Initial answers include 17 reference-correct answers, 11 non-null reference-wrong answers, and 13 null answers. Thirteen correct and nine incorrect answers

initially pass the closure gate. Table 9 preserves every policy outcome.

Disagreement-per-cost and the stateful validator match all 41 final correct/wrong-closure outcomes: each yields 14 correct and 11 incorrect closures. This sample tie is not population equivalence. Neither policy, nor any other evaluated comparator, corrects an initial reference-wrong answer. No policy changes an initially correct answer into an incorrect closure; the cheapest selector loses one initially correct closure through abstention. Additional checking can therefore increase the number of incorrect closures without repairing the underlying answer.

The proposed selector consumes 624,028 allocated tokens versus 597,739 for the validator: a 4.4% increase. The mean paired difference is 641 tokens per question, with capsule-bootstrap interval [13, 1449]. The mean tool-unit difference is 0.073, with interval [-0.024, 0.189]. These are recorded resource costs for this execution, not isolated causal effects of the selector.

All 41 first probes match between those two policies, as do 33 complete action sequences. The model supplies only one hypothesis for 27 questions; 79 of the selector’s 109 steps consequently or otherwise have zero predicted discrimination. Moreover, 59 of the validator’s 106 recommendations repeat an executed query and invoke the specified first-valid-probe fallback. This limits the original comparator. Among 118 groups of byte-identical formal model requests, 45 have differing response text despite fixed seed and temperature zero. The original runs are not deterministically identical inference replicas. A separate interface diagnostic below addresses the repeated-query fallback without replacing these frozen results.

12.5 Post hoc filtered-choice diagnostic

To remove the repeated-query fallback, an extra model call chooses an exact ID from the already filtered affordable, unexecuted probe list. Both policies receive this call, and its recommendation supplies the common tie break. Hypothesis generation, scoring, tools, evidence updates, and the final gate remain unchanged. Invalid choices leave the answer unresolved. This is a post hoc diagnostic on the same 41 cases, not another untouched evaluation.

The diagnostic makes 217 additional actual requests, consuming 857,480 tokens with no transport errors. Byte-identical within-case requests are reused across policies: 416 logical post-initial calls include 199 reuses. They are not independent model runs. Three policy trajectories encounter an invalid choice, which remains an unresolved outcome rather than triggering another fallback.

The validator closes 13 correct and 11 incorrect answers. The proposed selector closes 13 and 10. Neither corrects an initial wrong answer, and both lose one initially correct closure. The one outcome difference is abstention on the same incorrect proposed model name, not answer recovery. The paired false-termination difference is -2.4 percentage points with interval [-7.9, 0.0]; correct-closure difference is zero on every case. Allocated token totals are 919,775 and 919,655, respectively. The mean token difference is -2.9, with interval [-371.9, 447.9]. This repair does not establish a useful correction or cost advantage. Reliable candidate interpretations and outcome forecasts remain prerequisites for this selector.

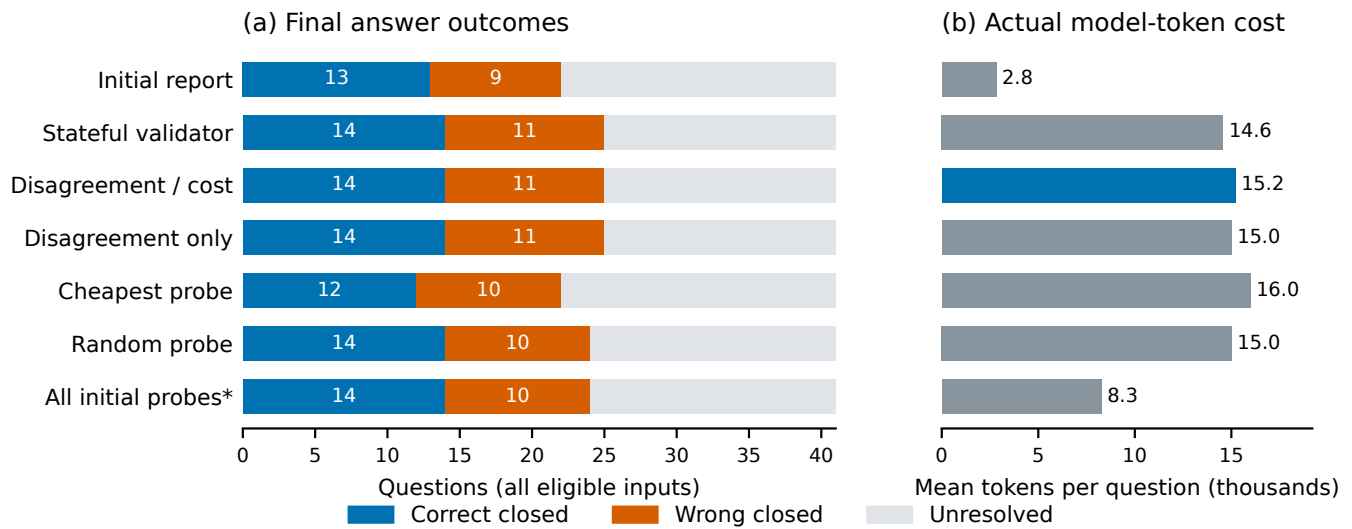


Figure 2: Frozen acquisition pilot. More inspection does not establish better answer correction. Token costs include the common initial response in each policy’s allocation; the actual total counts that response once. The starred all-initial-probes diagnostic executes the initial probe bank with its own budget and only one final answer update.

13 CBR Implementation and Replay Protocol

The extension was implemented in an isolated copy of the actual 30-stage research pipeline, preserving its original working tree and source hashes. Stages 24–28 perform execution-data, citation-grounding, logic, cross-modal, and computation-trace audits; Stage 29 aggregates reports and Stage 30 applies the trust gate. The original research-decision stage already writes structured PROCEED/REFINE/PIVOT outputs. Its runner records REFINE/PIVOT history and versions stages back to their configured restart point. These features do not provide historical input-bound decision replay.

The added receipt store records policy-code identity, frozen retry context, input digests, old action, predecessor and registered local outputs. Immutable receipt files are content addressed; the active history is atomically updated. A rollback first persists its intent and invalidated suffix, then moves registered outputs into a unique quarantine directory and writes the saved checkpoint. The next invocation completes a pending rollback before replay. This recovery mechanism is tested with a prepared rollback record. It does not cancel external jobs or restore unregistered state. Inputs must remain quiescent during validation, and the store uses a single-writer lock.

13.1 Integration boundary

The opt-in Stage-15 adapter captures the existing requirements-gate routing decision after the judge returns. Its canonical verdicts are PROCEED, PARTIAL, and REJECT; a remaining retry converts REJECT to REFINE. The original rule can select PROCEED after retry exhaustion despite unmet requirements. We preserve this behavior: replay correctness is not policy safety. Updating a registered verdict witness triggers counterfactual replay before the runner resumes. Upstream evidence-to-judge recomputation remains the producer’s responsibility. General free-form LLM decisions lack a frozen, deterministic replay adapter here and are outside the evaluated guarantee.

Ten focused behavior tests cover multiple flips, PIVOT parsing through the original parser, earliest rollback, unchanged-action refresh, unrelated changes, missing witnesses, exhausted replay budgets, policy changes, receipt tampering, path containment, persistence, and pending rollback recovery. A separate integration test executes the real Stage-15 function with four controlled judge/retry combinations, verifying both its structured output and captured receipt. These mocked-judge tests are software tests, not additional model calls. An initial adapter-enumeration mismatch was fixed after inspecting the real judge; the initial run is retained and the unchanged experiment was rerun under freshly frozen code hashes.

13.2 Executed branch trace

In an archived two-decision scenario, both decisions depend on a changed source. Their old actions are (REFINE, PROCEED). Replay leaves the first action unchanged but changes the second to REFINE. CBR retains the first output, quarantines the second, and persists a checkpoint identifying the second decision. Full replay produces the same result. The existing tracker correctly marks both dependent certificate checks false, but its interface does not revise the saved

PROCEED action. This is a controlled version change on a real artifact, not an observed natural research-policy error.

13.3 Accounting and limitations

There are 45 capsules, 405 event scenarios and 711 historical decisions per policy. Source changes and restorations produce 326 changed actions in 254 scenarios. Four policies yield 1,620 policy-event records. Independent code reconstructs file bytes and verifies action labels, receipt digests, checkpoint targets, output presence after rollback, and summary arithmetic. All cases are retained. Existing tracker work is 1,530 predicate evaluations; CBR’s 382 and full replay’s 416 are decision evaluations and should not be treated as interchangeable cost units. The 8.2% reduction is an observed work-count ratio, not a latency or population-effect claim. The experiment does not measure model tokens because no model is called.

The linear histories and version interventions are authored. Every event starts from the prior file state with newly recorded receipts; the experiment does not resume an autonomous research trajectory after each rollback. The registered expected versions are a contract oracle, not evidence that the underlying scientific claims are true. Complete read sets and frozen policies are assumptions. An augmented dependency tracker with action replay and the same rollback primitive can reproduce CBR; the current study does not demonstrate a new general incremental-computation algorithm.

13.4 Assistance and artifact

The author’s October 2 request specified auditing the seven added stages, implementing affected-decision replay with earliest-flip rollback, comparing four named policies, and updating the algorithm and artifact. Codex assisted with implementation, controlled experimental design, scoring and writing. The request, frozen experiment protocol, model-free test scope, source hashes, and reproduction commands are included in the artifact. No generated labels are described as independent human validation.

14 Preserved Protocol Details

14.1 Selective revalidation of changed inputs

We additionally implement a dependency tracker for versioned certificates. Let I_t be the checks directly bound to changed files. The invalidated set is

$$U_t = I_t \cup \text{Desc}_G(I_t). \quad (3)$$

The tracker clears cached decisions in U_t , re-evaluates them in topological order, and retains the other checks. For deterministic predicates with complete declared dependencies, this yields the same decisions as full re-audit: inputs outside U_t are unchanged, and induction over the affected order recomputes every changed dependency. Hidden inputs or changing verifiers invalidate this argument. This is standard incremental dependency tracking [3]; our experiment tests its certificate-lifetime role. The tracker is evaluated separately from the original acquisition loop, which continues to re-audit its full small graph.

Table 10: Fixed-retrieval SciFact operating points (%). Valid-claim recall means correct SUPPORTS/REFUTES resolution. Selective risk counts every wrong closed relation. Across the five saved runs, each LLM-containing policy has identical labels, so all operating-point sample standard deviations are 0.0; confidence rankings vary. Zero run variance does not imply zero sampling uncertainty.

Policy	Coverage $\uparrow$	NEI-FCR $\downarrow$	Valid-claim recall $\uparrow$	Macro-F1 $\uparrow$	Selective risk $\downarrow$
Direct NLI	58.1	33.8	56.0	59.5	40.6
LLM judge	72.6	46.9	64.6	60.2	45.1
Rule auditor	34.5	14.6	38.8	54.6	30.8
Risk-controlled NLI	27.7	9.2	34.0	53.2	24.5
Mondrian conformal (90%)	11.2	3.1	16.3	39.8	10.5
Global NLI–LLM consensus	30.4	10.8	36.8	54.9	25.2
Typed NLI only	41.0	12.3	52.2	65.3	21.6
TrustScientist semantic policy	34.8	8.5	46.9	63.5	16.9

15 SciFact: A Fixed-Retrieval Semantic Slice

15.1 Task, controls, and metrics

SciFact annotates whether scientific abstracts support, refute, or provide insufficient evidence for a claim [4]. The evaluator uses 339 unique development claim–document pairs: 138 SUPPORTS, 71 REFUTES, and 130 NEI (not enough information). Thresholds are chosen on 919 training pairs and frozen. In this setting, CLOSED means that the support/refutation relation is resolved. A correct REFUTES decision counts as correct closure while rejecting the claim’s asserted relation to the evidence. It does not grant publication acceptance.

Every policy receives the same top-ranked sentence from one frozen DeBERTa-v3 natural language inference (NLI) model. NLI scores entailment, contradiction, and neutrality between a sentence and a claim. NLI-only policies use one scoring pass; LLM policies can add one temperature-zero deepseek-chat call. The appendix records the exact NLI checkpoint. No policy obtains additional evidence. The selected sentence matches a gold rationale for only 56.5% of evidence-bearing pairs. Gold labels refer to the source evidence, so end-task errors can involve retrieval as well as relation prediction. The fixed evidence makes policy comparisons comparable but does not isolate these two causes causally.

Coverage is the fraction of all 339 pairs assigned SUPPORTS or REFUTES. NEI false-closure rate (NEI-FCR) is the fraction of the 130 gold-NEI pairs closed. *Valid-claim recall*, retained as a metric name for compatibility with the original evaluation, means correctly resolved gold SUPPORTS/REFUTES pairs divided by 209; it does not mean recall of true scientific propositions. Selective risk is the fraction of closed pairs whose relation is wrong, including both NEI closures and SUPPORTS/REFUTES reversals. Macro-F1 averages three-way class F1, mapping unresolved predictions to NEI. Macro-F1 is not a selective safety metric. These definitions make 98 correct closures among 118 total selections correspond to 46.9% valid-claim recall and 16.9% selective risk.

The policies include direct NLI, an LLM judge, a high-confidence rule auditor, a risk-controlled NLI rule, and 90% class-conditional Mondrian conformal prediction [1, 2]. Global NLI–LLM consensus uses a common relation threshold; typed NLI uses separate support

and refutation thresholds, 0.35 and 0.975. TrustScientist adds agreement with the LLM on the selected relation. Source admissibility is fixed, so this slice tests relation-specific selection and agreement. It does not separately exercise the four artifact-obligation types.

15.2 Utility, errors, and the consensus comparator

TrustScientist closes 118/339 pairs, including 98 correctly resolved evidence-bearing pairs. Its 20 wrong closures comprise 11 NEI cases and nine SUPPORTS/REFUTES reversals. Reporting only NEI-FCR would hide nearly half of these selected errors. Typed NLI attains higher Macro-F1 and recall; adding agreement lowers NEI closures from 16 to 11 while reducing correct relation closures from 109 to 98. This is an error–utility tradeoff, not dominance.

Global consensus correctly closes 77 evidence-bearing pairs and closes 14 NEI pairs. TrustScientist retains more correct decisions, but the difference of three NEI cases does not establish a safety advantage over consensus. Mondrian conformal prediction has the lowest NEI-FCR, four cases, while correctly closing only 34 evidence-bearing pairs. The supplied conformal configuration is a comparator; we make no transfer of its calibration guarantees to TrustScientist or post-action data.

The original matched-coverage analysis spans only 5–27.5% coverage, below TrustScientist’s 34.8% operating point. It integrates NEI-FCR, excluding relation reversals, and normalizes by the interval length. Its reported values, 0.0241 for TrustScientist and 0.0260 for global consensus, are numerically close. Clipped ranking scores also create ties whose order follows the saved records. The main paper retains the original matched-coverage curves; the appendix records their protocol. They do not establish a general risk–coverage advantage.

A post hoc rationale-hit diagnostic further separates the observed errors. On 118 evidence-bearing pairs whose sentence matches a gold rationale, TrustScientist makes 78 correct closures and one wrong relation; on 91 rationale misses it makes 20 correct closures and eight wrong relations. This stratification locates an association with evidence selection. It is not an oracle-retrieval experiment and does not assign errors causally.

All 11 NEI false closures involve agreement between typed NLI and the LLM on the same non-NEI relation. Ten also pass global

consensus with NLI scores above 0.98. One retrieved fragment contains a section cross-reference rather than a proposition and passes the lower support threshold. The examples show two limits: model agreement can preserve shared semantic mistakes, and a fixed retriever can supply evidence that cannot resolve the claim. An expert audit of those errors is distinct from their external SciFact labels; this study does not report a new human-validation experiment.

16 Recorded Result Visualizations

Figures 3–6 display the recorded cohorts with their original denominators. They add no new experiment. The figures distinguish branch consistency, certificate validity, benchmark-answer agreement and semantic relation resolution. These endpoints are not interchangeable measures of scientific correctness.

16.1 Reproduction and interpretation

The code and recorded evidence are available in the anonymous repository. The artifact contains the portable CBR implementation, its archived inputs and outputs, selected recorded cases, and the frozen inputs and plotting scripts for these figures. The CBR experiment runs offline on Linux with the Python standard library. The repository delivery uses split archives, unpacked by the first command below. The result figures require Matplotlib and NumPy; their data and scope captions are saved alongside the scripts. Model outputs are retained records, not independent human labels. The source manifest distinguishes the full offline CBR replay from re-computing statistics over previously recorded model outputs. New model inference and unrestricted autonomous scientific repair are not reproduced by the offline commands.

```
python unpack_artifact.py
python test_cbr.py
python run_branch_experiment.py prepare
python run_branch_experiment.py
python validate_branch_results.py
```

16.2 Maintenance and failure modes

CBR and full replay find all recorded earliest rollback points; the certificate-only tracker has no replay or rollback operation. The graph and flat dependency representations have identical certificate outcomes and predicate counts. Reduced work counts do not establish lower total latency, model cost or scientific verification cost. The acquisition panels retain the structured-selector tie and the semantic scope gate's loss of correct closures. Risk-qualified acquisition closes no case after the failed historical calibration.

At budget four, risk-qualified closures are 0/82. The same fresh-workspace tasks succeed in 19/20 cases, with 58/59 fresh answer matches in each saved run. Prepared compatibility fixes and the repeated task set limit the interpretation of this diagnostic.

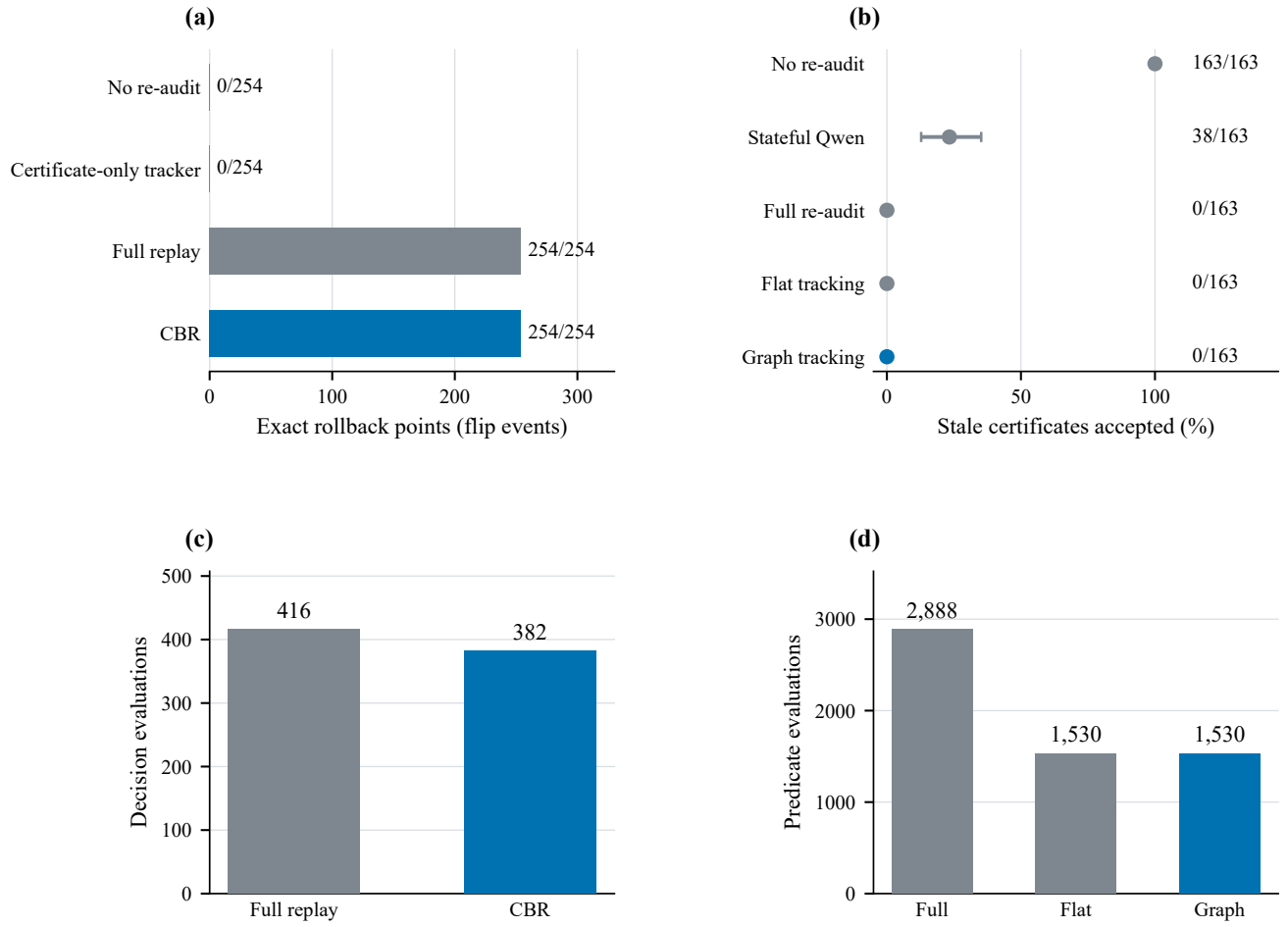


Figure 3: Controlled maintenance studies: 45 capsules and 405 update scenarios. CBR means Counterfactual Branch Revalidation. (a) Exact earliest rollback among 254 flip scenarios; the certificate-only tracker is the archived implementation without replay/rollback. (b) Stale certificate acceptance among 163 opportunities; whiskers are existing 45-capsule bootstrap 95% intervals. Degenerate empirical zero intervals are not population guarantees. Full, flat and graph retain 332/332 valid certificates; Qwen3-8B retains 327/332. (c) Decision evaluations for the two correct branch-replay policies. (d) Certificate predicate evaluations for equivalent deterministic policies. These separate work units exclude initialization and are not end-to-end costs. Histories are authored, changes scripted and dependencies supplied.

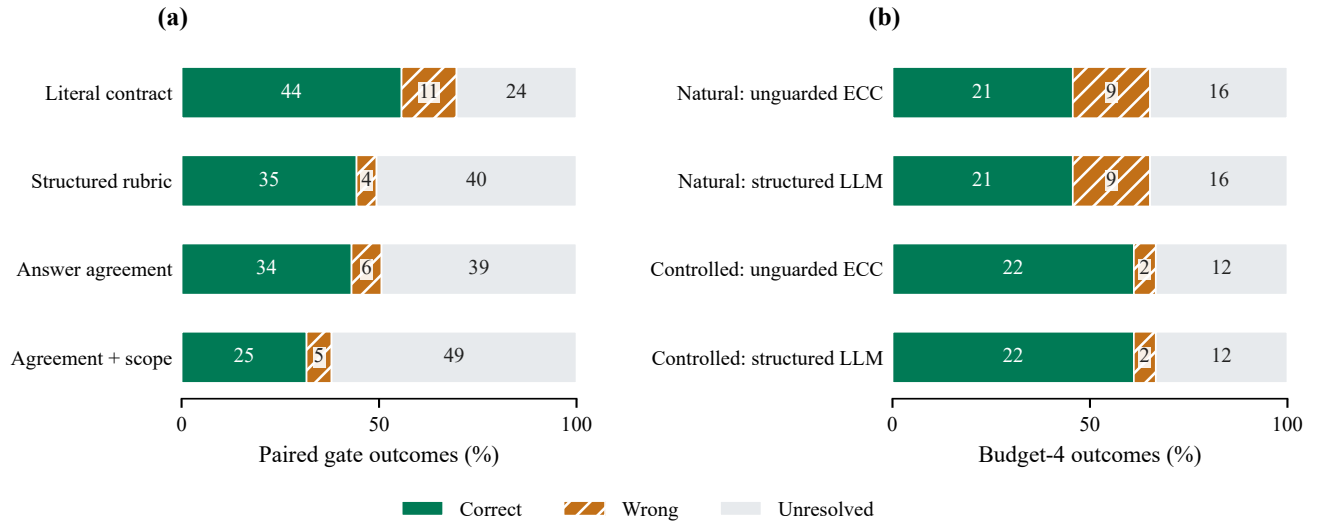


Figure 4: (a) Outcomes for 79 paired semantic-gate questions; correct/wrong mean benchmark-answer agreement, not expert scientific validation. Correct/Wrong count closed cases; Unresolved means not closed. (b) Budget-four existing-artifact acquisition on 46 natural reporting questions and 36 controlled states. ECC (Evidence-Closure Control) is unguarded here. Budget units are declared tool costs, not time or currency: rebinding/extraction costs one, search/fetch two, and broad refresh three. The structured large-language-model selector matches all 82 paired outcomes. The guarded controller closes no cases after failed calibration. Counts sum to each cohort denominator. Panels are not pooled; unrestricted autonomous recovery is untested.

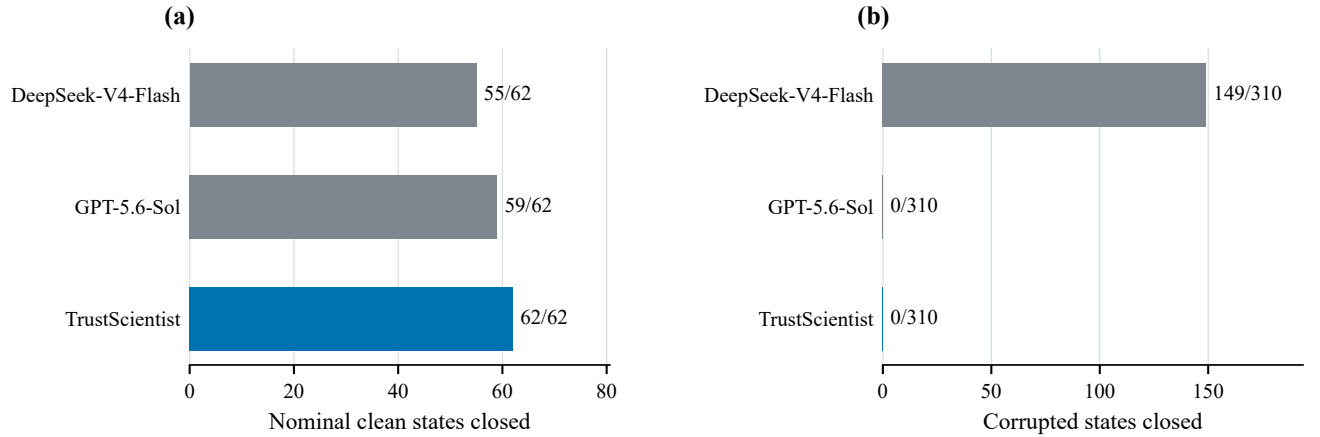


Figure 5: Matched supplied manifests: 62 nominal clean and 310 controlled corrupted states per policy. Clean closure is not semantic binding accuracy: three GPT nonclosures expose mismatches that TrustScientist accepts. Corrupted closure uses all 310 states, not a clean-closed conditional denominator. Models receive supplied validity flags, not raw artifacts. These are functional stress tests, not a natural-error or SOTA estimate.

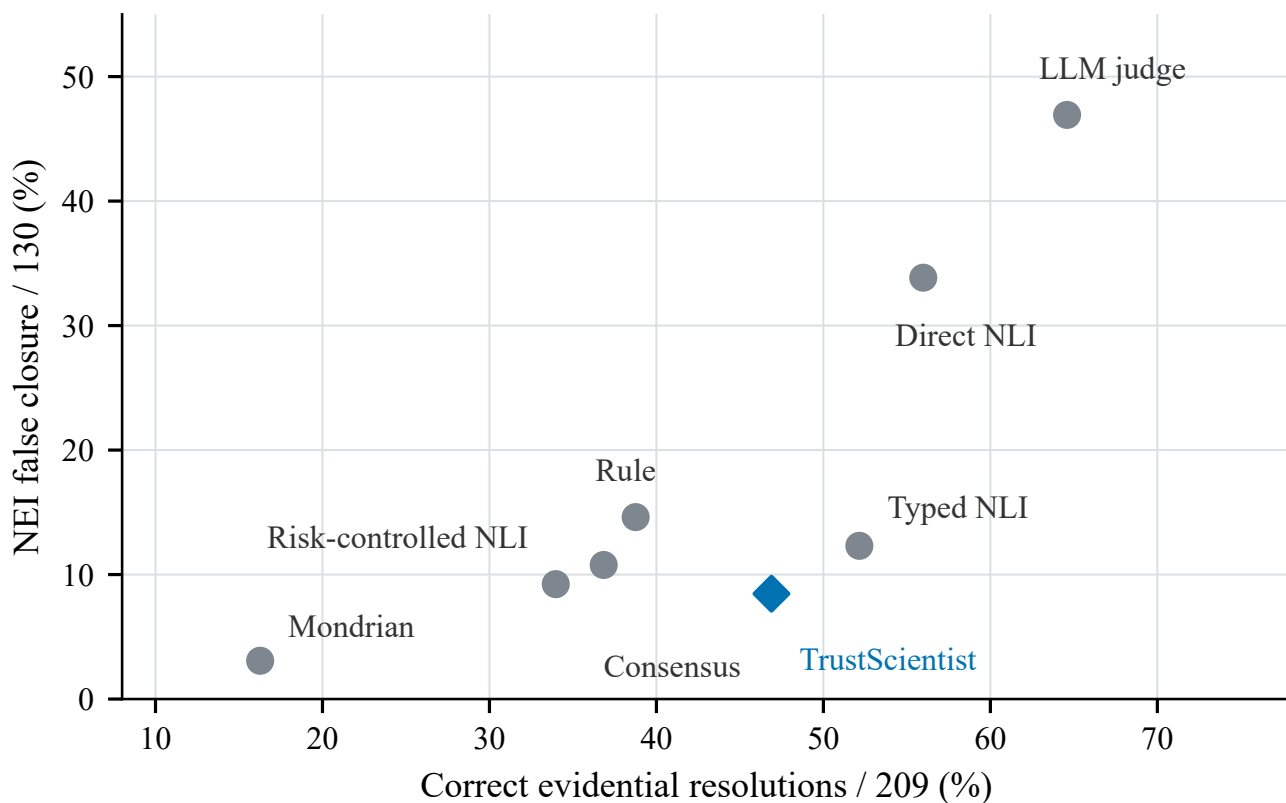


Figure 6: Fixed-retrieval SciFact: 339 pairs, with 209 evidence-bearing pairs and 130 NEI (Not Enough Information) controls. CLOSED resolves SUPPORTS/REFUTES, not publication acceptance. Mondrian uses class-conditional conformal prediction at $\alpha=0.10$ (90% nominal coverage); risk-controlled NLI uses a calibration risk target of 0.20. The six repeated policies use a representative saved run; their five-run aggregate operating-point metrics are identical. The two additional policies use single saved deterministic outputs. Methods have different closure coverage: points are not a matched-coverage frontier or connected curve. Lower NEI false closure can reduce useful resolutions.

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